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Preserving the Grand Design: A Ranking of ML Models for Long-Term Coherence in Symbolic Music Generation

This report provides a comprehensive analysis of the most effective neural and machine learning-based techniques developed between 2025 and the present for addressing long-term coherence and structural integrity problems in tokenized symbolic MIDI sequences. The primary criterion for evaluating effectiveness is the demonstrated ability to maintain long-range musical form, including thematic recurrence, section repetition, and hierarchical structure, rather than proxy metrics such as perplexity. The analysis synthesizes findings from recent academic papers and preprints, focusing exclusively on methods that operate on discrete tokenized representations of music derived from MIDI files. Each section details the mechanisms, evidence of effectiveness, and availability of source code for each identified technique, culminating in a ranked summary of the most impactful solutions.

Top-Tier Frameworks for Structural Integrity

The most significant advancements in generating structurally coherent music have emerged from frameworks that fundamentally shift the generation paradigm, moving beyond sequential autoregressive modeling. These top-tier approaches excel by either operating in a compressed structural latent space or by employing a systematic, theory-guided developmental process. They demonstrate superior performance because their core design philosophy is explicitly centered on planning and controlling musical form from the outset. This section analyzes two such leading frameworks: PhraseVAE/PhraseLDM, which utilizes a latent diffusion model for explicit structural conditioning, and Yin-Yang, which focuses on developing a central motif through a series of transformations and refinements. Both represent a departure from simply extending context windows and instead offer direct control over the piece's architecture.

The PhraseVAE/PhraseLDM framework introduces a paradigm shift in symbolic music generation by replacing autoregressive token-by-token generation with a latent diffusion

model operating on a compact, phrase-level representation [6] [9]. This approach directly addresses the inherent limitations of autoregressive models when tasked with maintaining consistency over very long sequences, which can span thousands of tokens [9]. The system's core innovation is its 'phrase-based generation paradigm' [9]. Instead of processing individual note events, it first compresses an entire musical piece into a much shorter sequence of latent vectors. Specifically, a Variational Autoencoder (PhraseVAE) encodes the music, where a 'phrase' is defined as all information for a single instrument within one bar [6]. This compression dramatically reduces the context length required by the generative model; for a 128-bar song, the number of elements to be processed drops from over 10,000 tokens to just 512 phrase-level latents [6]. The VAE employs a sophisticated multi-query compression mechanism using four learnable queries to attend to encoder hidden states, extracting different aspects of the musical information before projecting it into a final 64-dimensional vector [6] [9]. This method achieves high reconstruction fidelity, scoring a 99.0% F1 score for onset and pitch, which ensures that the compressed representation retains critical musical details [9].

Once the song is represented in this compact latent space, a Latent Diffusion Model (PhraseLDM) generates the entire multi-track composition in a single pass without any autoregressive components [6]. PhraseLDM is implemented as an encoder-only Transformer, specifically a DiT (Diffusion Transformer), which operates on the sequence of phrase-level latents [9]. The model's power lies in its controllability. It can be conditioned on optional inputs such as the desired song length and, crucially, the structural layout (e.g., `[intro - 8 bars, verse - 10 bars]`) [6]. This allows a user to directly prescribe the large-scale form of the generated piece, providing unprecedented control over its architecture. The implementation supports generating songs up to 128 bars long (approximately eight minutes at 64 BPM) and can produce a complete song in under 3.3 seconds on an NVIDIA A40 GPU, demonstrating its efficiency [6] [9]. The system was trained using the REMI-z grammar for tokenizing symbolic MIDI [6].

The effectiveness of PhraseVAE/PhraseLDM in preserving long-range structure is supported by compelling quantitative and qualitative evidence. Recognizing that standard metrics were insufficient, the authors introduced two novel evaluation tools based on a Self-Similarity Matrix (SSM) of bar-level latents. The first is the Section Recurrence Score (SRS), which quantifies the total length of non-major-diagonal lines in the SSM, representing recurring sections of music [9]. While real songs from the POP909 dataset achieve an average SRS of around 0.444, even the unconditional models learned some degree of structure planning, achieving an SRS of approximately 0.161 [6] [9]. The true

power of the framework is revealed through conditioning. Length conditioning improved the accuracy of generating songs of a target length (64 bars) from 15% to 95%. When structural layout conditioning was added, this accuracy reached 100%, proving that external prompts can effectively guide global form [9]. Furthermore, a double-blind listening test involving 15 professional participants demonstrated that PhraseVAE/PhraseLDM achieved competitive or superior scores in four key dimensions: Emotional Consistency (7.66), Melodic Fluency (7.90), Rhythmic Stability (7.30), and Structural Clarity (7.09). The improvements in Melodic Fluency and Structural Clarity were found to be statistically significant ($p < 0.05$), directly linking the framework's design to tangible improvements in perceived musical quality [4]. The project website provides audio examples and supplementary materials, offering verifiable proof of its capabilities [4].

The second top-tier framework, Yin-Yang, tackles long-term coherence through a different but equally powerful strategy: systematic motif development [17]. Instead of generating music freely, Yin-Yang begins with a musical motif and develops it into a longer piece while maintaining strong thematic identity. The framework consists of three independently trained transformer-based models: a phrase generator (PG), a phrase refiner (PR), and a phrase selector (PS) [17]. During generation, a corruption-refinement strategy is employed. Theory-inspired transformations—such as Fragmentation, Permutation, Inversion, Retrograde, Augmentation & Diminution, Reduction, and Masking—are applied to the initial motif to create variations. The PR model then takes these transformed phrases and refines them based on the preceding musical context, ensuring they fit coherently. Concurrently, the PS model selects the most suitable phrase from a pool of candidates generated by both the PG and PR, ensuring internal consistency within a given section [17]. This process creates a structured and homogenous piece that evolves from a single core idea.

The effectiveness of Yin-Yang in preserving long-range musical form is validated through novel objective metrics and subjective evaluation. To quantify the preservation of the initial motif, the authors introduce the Structural Derivation (SD) Score, which measures how smoothly the original theme manifests itself throughout the piece [17]. They also propose the Vendi Score, a diversity metric to ensure outputs are not overly repetitive [17]. Subjective listening tests showed significant improvements over strong baselines like the Music Transformer (MT) and Compound Word Transformer (CP). The Yin-Yang model achieved mean opinion scores of 3.52 for Coherence, 3.58 for Similarity, 3.59 for Structureness, and 3.41 for Overall quality, all significantly outperforming the baselines [17]. This demonstrates a clear enhancement in the listener's perception of a well-constructed, thematically unified composition. The models use the REMI sequence

encoding for symbolic music representation and were trained on monophonic folk song datasets [17]. The entire system is available via a public repository, allowing for verification and further research [17]. The Yin-Yang framework's strength lies in its focused, intentional approach to development, making it exceptionally effective for creating pieces with deep thematic coherence, a cornerstone of long-term musical structure.

Advanced Architectural Innovations for Long-Context Modeling

Beyond fundamental shifts in generation paradigms, another category of highly effective techniques involves advanced architectural innovations designed to capture long-range dependencies more efficiently and effectively. These methods build upon established architectures like Transformers but introduce novel mechanisms to manage memory, attention, and computational complexity. They address the core challenge of maintaining coherence across thousands of tokens by re-engineering how the model processes and retains information over long distances. This section examines several prominent examples, including Depth-Structured Music Recurrence (DSMR), which uses a sophisticated memory scheduling scheme in a recurrent Transformer; PerceiverS, which improves upon the Perceiver AR architecture with better training dynamics and multi-scale attention; and hybrid models like MusicMamba and SMDIM, which combine the strengths of Transformers with those of State Space Models (SSMs).

Depth-Structured Music Recurrence (DSMR) is a recurrent Transformer architecture specifically designed for full-piece symbolic music modeling by extending context beyond fixed-length excerpts [14]. Its primary innovation is the "layer-wise memory-horizon schedule." In a traditional Transformer, every layer has access to the same context window. DSMR breaks this uniformity by allocating a budget for recurrent Key-Value (KV) states and distributing it across different layers. Lower layers are assigned a long history window, enabling them to specialize in capturing long-range dependencies such as motif repetition and global form. Higher layers receive progressively shorter windows, forcing them to focus on shorter-term local context and fine-grained details [14]. This creates depth-dependent temporal receptive fields, allowing the network to process entire musical pieces in a single left-to-right pass without truncation. By carrying recurrent states forward from one segment to the next, every token in the piece can condition on the long-range causal history spanning the entire work, a critical feature for maintaining

coherence [14]. The authors found that a "two-scale DSMR" schedule, which allocates a long history window to lower layers and a short window to the rest, served as a strong default for this task [14]. The model was evaluated on the MAESTRO piano performance dataset, using tokenized symbolic event sequences. It achieved a Best Validation Perplexity (PPL) of 5.96, outperforming other advanced methods like gate-guided selective retention (6.53) and a Perceiver-AR-like reference (6.54). Crucially, DSMR achieved this performance using over 50% less GPU memory (6.3 GB) compared to a full-attention reference model (15.5 GB), highlighting its efficiency [14]. While the provided documents do not contain a direct link to a code repository, the described mechanism represents a powerful and efficient solution to the long-context problem.

PerceiverS builds upon the Perceiver AR model but introduces two key innovations to improve its ability to generate long-term expressive music [15] [16]. First, it addresses a critical flaw in Perceiver AR's training process called "Effective Segmentation." In the original model, tokens could inadvertently "peek" at unintended future tokens due to issues with causal masking during training. PerceiverS solves this by progressively increasing the context length from short segments up to the maximum input length (32,768 tokens) during training. This aligns the training phase more closely with auto-regressive generation and ensures that the model learns to generate tokens correctly, preventing it from relying on forbidden future information [15] [16]. Second, to combat the tendency of models relying on ultra-long contexts to generate repetitive segments, PerceiverS implements a "Multi-Scale cross-attention mechanism" [15]. This involves multiple layers of cross-attention with varying scales of attending length. One layer attends to all previous tokens, while a subsequent layer applies a scale mask that only allows attention to a shorter, recent context window (e.g., the last 1,024 tokens). The outputs are combined in a cascade, forcing the model to balance its focus between long-range structural dependencies and short-range variations [16]. Experiments on the Maestro dataset, which contains 1,251 piano pieces, demonstrated significant improvements. The combination of Effective Segmentation and Multi-Scale attention resulted in an average 40% improvement in Overlap Area (OA) when measured against the original training dataset [16]. Human evaluations confirmed these gains, with PerceiverS achieving an average score of 3.29 out of 5 for overall musical quality, compared to the baseline's 2.00. Assessors rated PerceiverS higher in tempo coherence (3.82 vs 2.86), motif presence (2.82 vs 2.14), and harmonic progression (2.79 vs 1.64) [16]. The project website offers demos and generated samples, providing a platform for auditory assessment of its output quality [15].

Hybrid architectures like MusicMamba and SMDIM represent another frontier in tackling long-sequence generation by combining the strengths of different neural building blocks.

MusicMamba integrates the long-range dependency modeling of Mamba Blocks with the global structure capturing ability of Transformer Blocks [3]. It introduces a Bidirectional Mamba Fusion Layer that processes features from both directions separately, applies self-attention to each direction, concatenates them, and passes the result through a linear layer to create a comprehensive representation for generation [3]. This fusion allows the model to simultaneously capture both local melodic details and overall structural patterns like phrases and repetitions. The model also introduces a new tokenization scheme, REMI-M, which extends the standard REMI representation by adding events to explicitly annotate and generate modal changes, which is particularly useful for genres with complex tonal structures like Chinese traditional music [3]. SMDIM (Symbolic Music Diffusion with Mamba) is a related framework that combines structured state space models (SSMs) with a hybrid refinement scheme [4]. It uses a hierarchical MFA block (Mamba → FeedForward → Attention) to balance computational efficiency with the preservation of fine-grained musical detail. Both models leverage structured tokenizations like REMI-M and REMI to enable the model to learn rhythmic stability, phrasing, and harmonic structure [3] [4].

The effectiveness of these hybrid models is clearly demonstrated through both objective and subjective evaluations. MusicMamba was shown to outperform baselines like MusicTransformer and MelodyT5 in generating Chinese traditional music. Subjective listening tests yielded high scores for coherence (7.91), richness (7.72), and style (8.26). Objective metrics showed that over 70% of generated music exhibited a detectable modal structure [3]. Similarly, SMDIM achieved a competitive Average Overlapping Area (OA) score of 0.850 on the Chinese Tradition dataset, outperforming variants with only Transformers or only Mambas. Double-blind listening tests demonstrated that SMDIM achieves competitive or superior scores in four key dimensions, with statistically significant improvements in Melodic Fluency and Structural Clarity ($p < 0.05$) [4]. The SMDIM project webpage provides audio examples and supplementary materials, offering verifiable proof of its capabilities [4]. These hybrid models are powerful because their architectural fusion allows them to capture a wider range of musical phenomena than a single-architecture model, balancing the efficiency of SSMs with the global modeling capacity of Transformers.

Strategic and Methodological Approaches to Form Generation

In addition to architectural innovations, several effective techniques rely on strategic and methodological approaches that guide the generation process towards greater structural integrity. These methods often involve injecting prior knowledge about musical structure into the model's functioning, either by modifying core components like positional encodings or by designing novel training and inference procedures. They are less about changing the underlying model backbone and more about teaching it *how* to perceive and generate structure. This section explores two distinct but powerful strategies: Fast Structure-Informed Positional Encoding (F-StrIPE), which enhances Transformers by embedding structural information directly into their positional representations, and MIDI-RWKV, which leverages a specialized linear architecture for long-context awareness and introduces a parameter-efficient finetuning method called "state tuning."

F-StrIPE (Fast Structure-Informed Positional Encoding) is a lightweight yet effective technique designed to enhance standard Transformer models for symbolic music generation [20]. Its core mechanism is to replace the standard linear positional indices (1, 2, 3, 4, ...) with vectorial structural labels at different resolutions, such as note, chord, and phrase levels. This creates a non-linear grid that explicitly encodes hierarchical musical information directly into the model's attention mechanism. By doing so, F-StrIPE injects prior knowledge about musical organization into the model without requiring any changes to the training pipeline or the underlying architecture [20]. The method generalizes Stochastic Positional Encoding (SPE) by using Random Fourier Features (RFF), which provides a noiseless estimate of the positional matrix compared to SPE's stochastic approximation, and it maintains linear complexity in sequence length, making it suitable for very long sequences [20]. The technique is publicly available via a website containing music samples and code at `bit.ly/faststructurepe` [20].

The effectiveness of F-StrIPE is demonstrated through empirical evaluation on melody harmonization tasks using the POP909 dataset [20]. The results show that F-StrIPE significantly improves performance over standard positional encodings (NoPE) and SPE. The best-performing variant, F-StrIPE:C, which uses only chord-level structure information, outperformed all others on metrics like Chroma Similarity (CS) and Grooving pattern Similarity (GS). Notably, the improvements were more pronounced when tested on longer sequences (64 bars versus 16 bars), indicating that the injected structural priors become increasingly valuable as the context window expands [20]. This suggests that F-StrIPE helps the model maintain coherence over longer spans by giving it

an explicit map of the musical hierarchy. While F-StrIPE is a supporting technique rather than a complete generative system, its elegance and low computational cost make it a valuable tool for improving the structural perception of any underlying Transformer-based music generation model.

MIDI-RWKV presents another innovative approach by replacing the Transformer decoder with the RWKV-7 linear architecture [5]. This choice is motivated by the RWKV model's known ability to handle long contexts efficiently due to its constant hidden state size, which allows for an unlimited context window during inference—a key feature for capturing long-range dependencies [5]. The model uses the REMI encoding for multi-track MIDI files and further enhances its vocabulary by applying Byte-Pair Encoding (BPE), expanding it from 663 to 16,000 tokens to reduce sequence length [5]. To improve performance on long-context tasks, the authors developed a novel prompt format called "single-section infilling," which allows the model to concentrate its attention on a single target section by masking all others, unlike prior methods that masked all potential sections simultaneously [5]. The most unique contribution of MIDI-RWKV is a novel finetuning method called "state tuning." Instead of updating the millions of parameters in the model weights, state tuning optimizes only the initial state vectors for each layer using cross-entropy loss. This requires significantly fewer parameters (e.g., ~294k vs. millions for LoRA) and can be performed very quickly, in minutes, enabling rapid personalization [5].

The effectiveness of MIDI-RWKV is demonstrated through rigorous evaluation. On objective metrics for long-context tasks like 8-bar infillings, MIDI-RWKV demonstrated superior performance over MIDI-Mistral, holding long-context coherence significantly better across all metrics, including Content Preservation (CP), Groove Similarity (GS), and Pitch Class Histogram Entropy difference (PCHE) [5]. Most importantly, subjective listening tests revealed that a state-tuned version of the model was significantly preferred over both the base model and a LoRA-finetuned model. Participants ranked the state-tuned output highest, indicating improved musical quality and coherence [5]. The model's architecture, based on the RWKV-7 linear model, was trained on the GigaMIDI dataset, consisting of 1.05 million MIDI files, and the code and model weights are publicly available on GitHub [5]. MIDI-RWKV's strength lies in its combination of a highly efficient architecture for long contexts with a novel, practical method for personalization, making it a powerful tool for computer-assisted composition that prioritizes long-term coherence [5].

Emerging Paradigms and Future Directions

While many of the most mature techniques detailed in the provided sources focus on refining existing architectures or introducing specific modules, the research landscape in 2025 also shows the emergence of entirely new conceptual paradigms aimed at overcoming the challenges of long-sequence music generation. These methods are often presented as preprints or in early-stage publications, suggesting they are at the forefront of current research. Although they may lack the extensive evaluation and code availability of more established models, their titles and abstracts reveal novel approaches that could define the next generation of structural music generation. This section highlights two such emerging paradigms: "Anchored Cyclic Generation," which proposes a novel method for generating long sequences, and "SymphonyGen," which focuses on hierarchical orchestration as a means of managing large-scale musical structure.

One preprint titled 'Anchored Cyclic Generation: A Novel Paradigm for Long-Sequence Symbolic Music Generation' by Boyu Cao et al. points toward a new approach to the fundamental problem of generating long, coherent musical passages [2]. The title itself suggests a departure from purely autoregressive or latent diffusion methods. "Anchored Cyclic Generation" implies a process where certain points in the sequence (the "anchors") are fixed or strongly influenced, and the model generates the content between them in a cyclic or iterative fashion. This could be a way to enforce structural constraints, such as ensuring that a specific melodic phrase returns at a designated point in the piece, thereby directly combating the common issue of structural drift in long compositions. While the provided context lacks the technical details of the mechanism, its identification as a novel paradigm targeting long-sequence generation signals a significant research direction [2]. Such a method would need to be rigorously evaluated against existing benchmarks to prove its superiority, but its conceptual framing suggests a potential solution to the coherence problem that is distinct from current mainstream approaches.

Another preprint, 'SymphonyGen: 3D Hierarchical Orchestral Generation with Controllable Harmony Skeleton,' by Xuzheng He et al., offers a glimpse into a future where generation models are designed to handle the immense complexity of large-scale, multi-instrumental works [2]. The title indicates a focus on "3D hierarchical orchestration," which likely refers to organizing music not just in time (the 1D timeline) and across instruments (the 2nd dimension), but also across different layers of abstraction or instrumentation density (the 3rd dimension). The mention of "controllable harmony skeleton" suggests a method for generating a foundational harmonic structure first, which then guides the filling in of melodic and instrumental details. This approach mirrors compositional practices in classical and film music, where a composer might

sketch out the chord progression and overall form before orchestrating it. By managing structure at a high level of abstraction (the harmony skeleton) and then generating the concrete musical texture, SymphonyGen aims to impose a robust, large-scale form on the final piece [2]. Like Anchored Cyclic Generation, the specifics of the model architecture and its performance metrics are not detailed in the provided sources. However, its focus on hierarchical orchestration implies a sophisticated set of techniques for managing the interplay between different musical elements over extended durations, which is a hallmark of long-term coherence [2]. These emerging paradigms, while still in their early stages, represent a crucial evolution in the field, moving towards more sophisticated, multi-layered, and conceptually richer models of musical generation.

Comparative Analysis and Code Availability

To provide a clear overview of the most effective techniques, this section presents a comparative analysis summarizing the key characteristics, mechanisms, and verifiable code availability for each method discussed. This table serves as a quick-reference guide, consolidating the detailed information from the preceding sections. The ranking is based on the demonstrated ability to preserve long-range musical form, considering both the novelty of the approach and the strength of the evidence provided in the source materials. Effectiveness is determined by a combination of factors, including the introduction of novel evaluation metrics, superior performance on relevant quantitative metrics, and positive results from subjective listening tests.

Technique/ Framework	Core Mechanism	Tokenization Scheme(s)	Evidence of Effectiveness	Verifiable Code/Repository
PhraseVAE / PhraseLDM	Latent diffusion model operating on a compact, phrase-level latent space. Allows explicit conditioning on structural layouts (e.g., intro, verse). [6] [9]	REMI-z [6]	Introduced novel metrics: Section Recurrence Score (SRS) and Bar-level SSM. Unconditional models learn structure (SRS ~0.16). Structural conditioning increases length accuracy to 100%. Subjective tests show statistically significant improvement in Melodic Fluency and Structural Clarity ($p < 0.05$). [9]	Public repository and demos available at: `https://www.oulongshen.xyz/midi_ldm` [6] [9]
Yin-Yang	Motif-driven generation using a corruption-refinement strategy with three independently trained transformers (generator, refiner, selector). Applies theory-inspired transformations to develop a core motif. [17]	REMI [17]	Introduced novel metrics: Structural Derivation (SD) Score and Vendi Score. Subjective tests show significant preference over baselines (Music Transformer, CP) in Coherence (3.52), Similarity (3.58), and Structureness (3.59). [17]	Public repository available at: `https://github.com/keshavbhandari/yinyang` [17]
Depth-Structured Music Recurrence (DSMR)	Recurrent Transformer with a "layer-wise memory-horizon schedule" that allocates longer history windows to lower layers to specialize in long-range dependencies. Processes full pieces in one pass. [14]	Tokenized symbolic event sequences (derived from MIDI) [14]	Achieved a Best Validation Perplexity (PPL) of 5.96 on the MAESTRO dataset, outperforming other advanced methods while using over 50% less GPU memory. [14]	Information not available in provided sources.
PerceiverS	Improves Perceiver AR with "Effective Segmentation" to fix training-to-inference mismatch and a "Multi-Scale cross-attention mechanism" to balance long-range and short-range context. [15] [16]	Not Available	Combination of techniques led to a 40% improvement in Overlap Area (OA). Human evaluation showed significant gains in overall musical quality (3.29 vs 2.00 baseline), tempo coherence, and motif presence. [16]	Project website with demos available at: `https://perceivers.github.io` [15]
MusicMamba / SMDIM	Hybrid architectures combining Transformer blocks (for global structure) with Mamba/ State Space Model (SSM) blocks (for local detail and efficiency). [3] [4]	REMI-M (MusicMamba), REMI (SMDIM) [3] [4]	MusicMamba: High Mode Consistency (>70%) and strong subjective scores for Coherence (7.91) and Style (8.26). SMDIM: Achieved OA score of 0.850 and statistically significant improvement in Melodic Fluency and Structural Clarity ($p < 0.05$). [3] [4]	SMDIM project webpage with audio examples: `https://3328702107.github.io/smdim-music/` [4]. MusicMamba code info not available.
F-StrIPE	Lightweight technique enhancing Transformers	Not Available	Significantly improves performance on melody	Code and samples available at: `bit.ly/faststructurepe` [20]

Technique/ Framework	Core Mechanism	Tokenization Scheme(s)	Evidence of Effectiveness	Verifiable Code/Repository
	by replacing linear positional indices with vectorial structural labels (note, chord, phrase) at multiple resolutions. Injects structural knowledge into the PE module. [20]		harmonization (POP909 dataset), especially on longer sequences (64 bars). Metrics like Chroma Similarity (CS) and Grooving pattern Similarity (GS) show marked improvement. [20]	
MIDI-RWKV	Uses the RWKV-7 linear architecture for long-context awareness. Introduces "state tuning," a parameter-efficient finetuning method that optimizes initial state vectors. [5]	REMI with Byte-Pair Encoding (BPE) [5]	Superior performance on long-context infilling tasks (8-bar) over MIDI-Mistral. Subjective tests show state-tuned version is significantly preferred for coherence and quality. Unlimited context window aids long-range dependency capture. [5]	Public repository and model weights available at: `https://github.com/christianazinn/MIDI-RWKV` [5]

This comparative view underscores the diversity of successful strategies. The top-ranked methods, PhraseVAE/PhraseLDM and Yin-Yang, succeed by offering direct control over musical form. The next tier of methods, including DSMR, PerceiverS, and the hybrid models, excel by innovating on the architecture itself to better handle long-range dependencies. Finally, techniques like F-StrIPE and MIDI-RWKV provide powerful, often computationally efficient, enhancements to existing frameworks, demonstrating that effectiveness can come from both grand architectural visions and clever, targeted modifications. The widespread availability of code for many of these cutting-edge models is a significant boon to the research community, facilitating reproducibility and further innovation.

Synthesis and Final Ranking of Effective Techniques

Based on a comprehensive analysis of the provided research from 2025 to the present, a clear hierarchy of techniques for alleviating long-term coherence and structure problems in tokenized symbolic MIDI sequences emerges. The ranking is strictly based on the demonstrated ability to preserve long-range musical form, incorporating the novelty of the approach, the strength of the evidence (quantitative and qualitative), and the level of structural control offered. The most effective solutions are those that move beyond simple sequence extension and instead employ strategies that actively plan, condition, or guide the generation of musical structure.

First Place: PhraseVAE / PhraseLDM (Latent Diffusion Framework) This framework is ranked #1 because its entire design philosophy is centered on solving the long-term coherence problem. By abstracting music into a compact, phrase-level latent space, it bypasses the exponential growth of errors in autoregressive models over long sequences [6] [9]. Its most significant contribution is the ability to provide explicit structural conditioning, allowing a user to dictate the form of the piece (e.g., intro, verse, chorus) [6]. This direct control over global form is unparalleled among the reviewed methods. The introduction of novel evaluation metrics like the Section Recurrence Score (SRS) and Bar-level Self-Similarity Matrix (BarSSM) provides verifiable, domain-relevant evidence of its success, showing that the model not only follows instructions but also learns inherent structure planning on its own [9]. The statistically significant improvements in subjective listening tests for Melodic Fluency and Structural Clarity further cement its position as the most effective technique for ensuring large-scale coherence [4]. The availability of its codebase makes it a practical and accessible solution [6].

Second Place: Yin-Yang (Motif-Driven Generation) The Yin-Yang framework secures the #2 spot by offering an exceptionally effective solution for a specific but critical aspect of long-term structure: thematic coherence. Its methodology, which begins with a single motif and systematically develops it through a series of theory-inspired transformations and context-aware refinements, ensures a deep sense of unity and purpose throughout a piece [17]. This focused approach guarantees that the music will return to and evolve its core ideas, a hallmark of well-structured compositions. The creation of the Structural Derivation (SD) Score provides a novel, verifiable metric to quantify this success, moving beyond generic quality assessments [17]. The significant preference shown in subjective listening tests for its coherence, similarity, and structureness compared to strong baselines confirms its practical effectiveness [17]. While perhaps less flexible than PhraseVAE/PhraseLDM for free-form composition, its methodical and intentional approach to development makes it a powerhouse for creating thematically integrated music.

Third Place: Depth-Structured Music Recurrence (DSMR) Ranked #3, DSMR is a powerful architectural innovation that directly addresses the computational and representational challenges of long-sequence modeling [14]. Its "layer-wise memory-horizon schedule" is a clever and efficient solution that enables a recurrent Transformer to process entire musical pieces in a single pass, granting every token access to the causal history of the whole work [14]. This is fundamental to maintaining coherence. Its effectiveness is demonstrated by achieving state-of-the-art validation perplexity on a major benchmark while being significantly more memory-efficient than comparable models [14]. While its effectiveness is more implicit—as a superior engine for processing

long contexts—it provides a robust foundation upon which structural integrity can be built. It stands out as a highly efficient and performant solution for anyone seeking to apply recurrent models to full-piece music generation.

Fourth Place: PerceiverS (Effective Segmentation & Multi-Scale Attention) PerceiverS ranks #4 as a highly effective set of improvements applied to an existing, capable architecture [15] [16]. Its two core innovations—the "Effective Segmentation" training strategy and the "Multi-Scale cross-attention mechanism"—directly solve known weaknesses of the Perceiver AR model, namely the training-to-inference mismatch and the tendency to generate repetitive content [15] [16]. The resulting 40% improvement in Overlap Area (OA) and significant gains in subjective human evaluations for overall quality, tempo coherence, and motif presence provide strong evidence of its effectiveness [16]. It represents a pragmatic and powerful path to enhancing the capabilities of advanced architectures, making it a very strong contender.

Fifth Place: MusicMamba / SMDIM (Hybrid Architectures) These hybrid models rank #5 for their innovative fusion of Transformer and State Space Model (SSM) architectures [3] [4]. This combination allows them to leverage the global modeling strength of Transformers and the efficiency and local detail handling of SSMs, creating a balanced and powerful generative engine. Their effectiveness is clearly demonstrated through both objective metrics (high mode consistency, competitive OA scores) and subjective tests showing significant improvements in Melodic Fluency and Structural Clarity [3] [4]. They represent a significant step forward in architectural design, though their primary innovation is the fusion of components rather than a dedicated structural generation strategy like the top-ranked methods.

Sixth Place: F-StrIPE (Fast Structure-Informed Positional Encoding) Ranked #6, F-StrIPE is an elegant and computationally inexpensive technique that enhances any standard Transformer by injecting structural knowledge directly into its positional encoding module [20]. Its effectiveness is proven by its ability to significantly improve performance on longer sequences, helping the model maintain coherence by providing an explicit map of the musical hierarchy [20]. While it is a supporting technique rather than a complete generative system, its value is immense. It acts as a force multiplier, improving the structural perception of any underlying model. Its availability via a public link makes it a widely accessible tool for researchers and developers.

Seventh Place: MIDI-RWKV (RWKV Linear Architecture) Ranking #7, MIDI-RWKV excels at handling long contexts due to its efficient RWKV-7 architecture and unlimited context window [5]. Its introduction of "state tuning" is a valuable contribution, offering a

fast and parameter-efficient method for personalizing the model [5]. It demonstrates superior performance on long-context tasks and receives high marks in subjective tests for coherence and quality. It is a very strong performer, particularly noted for its efficiency and controllability, but its application is more general-purpose for long-context tasks rather than being specifically tailored to enforcing complex musical forms.

In conclusion, the most effective techniques for ensuring long-term musical coherence are those that prioritize structural control. The PhraseVAE/PhraseLDM framework sets the highest bar by offering direct, explicit control over a piece's form. The Yin-Yang framework demonstrates that a focused, developmental approach to a single motif is another highly effective path. Other powerful methods, including DSMR, PerceiverS, and hybrid models, advance the field by innovating on the underlying architecture to better process and retain long-range information. The consistent use of standardized, semantically-rich tokenization schemes like REMI across these models provides a common foundation that facilitates the learning of musical grammar and structure [4] [5] [17].

Reference

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